

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	03 October 2026
Team ID	SWTID-2026-1387
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The deliverable details the system architecture and information flow for automating standard laptop procurement using ServiceNow Flow Designer under the Nan Mudhalvan initiative.

Workflow & System Architecture Steps:

1. User Request Initiation: Employee accesses the Service Catalog via ServiceNow / Service Portal and submits a standard laptop request with auto-populated user and department metadata.
2. Automated Approval Routing: Flow Designer triggers conditional approval workflows to Reporting Manager and IT Procurement Lead via Email and Service Portal notifications.
3. Catalog Task Creation & Assignment: Upon approval, Flow Designer executes the flow "Standard Laptop Task", creates a Catalog Task with short description "Laptop need to Configured", sets Assignment Group to "Hardware", and sets Approval status to "Approved".
4. Vendor PO Generation & Fulfillment: Automated REST API call sends the order payload directly to the external hardware vendor endpoint for fulfillment.
5. Delivery, OS Flashing & Setup: Desktop Support Engineer receives physical laptop, flashes base OS image, installs required software, and dispatches device to employee.
6. Asset Lifecycle & CMDB Sync: ServiceNow CMDB automatically updates hardware state from "In Stock" to "In Use" and triggers an automated satisfaction survey.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Service Portal / Service Catalog interface allowing users to submit laptop orders and track request status	HTML, CSS, JavaScript, AngularJS, ServiceNow Service Portal
2.	Application Logic-1	Flow Designer automation flow ("Standard Laptop Task") to create catalog task ("Laptop need to Configured") assigned to "Hardware" group	ServiceNow Flow Designer
3.	Application Logic-2	Workflow approval logic for multi-tier manager and IT procurement lead validations	ServiceNow Workflow Engine / Approval Rules
4.	Application Logic-3	IntegrationHub REST API module to send purchase order payloads to hardware vendor endpoints	ServiceNow IntegrationHub / REST Web Services
5.	Database	Relational database storing catalog items, service requests, user metadata, and task records	MySQL / ServiceNow Relational Database Engine
6.	Cloud Database	Cloud-hosted CMDB data store for tracking hardware asset states and lifecycle transitions	ServiceNow Cloud Instance CMDB
7.	File Storage	Storage for purchase receipts, approval signatures, software configuration scripts, and documentation	ServiceNow System Attachment Engine / Cloud File Storage
8.	External API-1	Vendor fulfillment REST API endpoint for automated order processing	Vendor REST API / OAuth 2.0 Webhook
9.	External API-2	Email & messaging webhook integration for approval alerts and tracking updates	SMTP / Microsoft Teams Webhook API / Twilio SMS
10.	Machine Learning Model	Predictive Intelligence model for request anomaly detection and delivery timeline estimation	ServiceNow Predictive Intelligence ML Engine
11.	Infrastructure (Server / Cloud)	High-availability cloud platform hosting ServiceNow SaaS instances	Cloud SaaS / ServiceNow Multi-Instance Architecture

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frontend and integration scripting frameworks for user experience and API communication	JavaScript (ES6+), AngularJS, REST Web APIs

S.No	Characteristics	Description	Technology
2.	Security Implementations	Role-Based Access Control (RBAC), end-to-end TLS 1.3 encryption, and ACL security rules	OAuth 2.0, SSL/TLS, ServiceNow ACLs, IAM Controls
3.	Scalable Architecture	Cloud SaaS multi-tenant / multi-instance architecture supporting concurrent workflow executions	ServiceNow Flow Designer Engine, Microservices
4.	Availability	High availability deployment with failover nodes and load balancers ensuring continuous uptime	ServiceNow High Availability (HA) Architecture, Load Balancers
5.	Performance	Asynchronous execution of flow triggers, cached catalog schemas, and optimized database indexing	Redis Caching, Asynchronous Event Queue, CDN

References:

<https://c4model.com/>

<https://developer.servicenow.com/>

<https://docs.servicenow.com/bundle/tokyo-application-development/page/administer/flow-designer/concept/flow-designer.html>

<https://portal.naanmudhalvan.tn.gov.in/>